

ARITRO DUTTA

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Work Rights: Full Working Rights in Australia (Subclass 485 Graduate Visa - Valid to Jan 2028, No Restrictions, No Employer Sponsorship Required)

PROFESSIONAL SUMMARY

Software Engineering (Honours) graduate from Monash University with production experience across AI development, full stack engineering and data pipelines. Backed by two structured industry placements at QEERI.AI and CSIRO and one year of hands-on engineering team experience at Monash Uncrewed Aerial Systems, delivering real outcomes in production environments before graduating. Comfortable owning work end to end, from shaping the approach through to shipping and fixing what breaks. **Do not require any employer sponsorship and am actively pursuing permanent residency at my own expense. Available immediately.**

EDUCATION

Bachelor of Software Engineering (Honours) | Monash University

Dec 2025

Melbourne, VIC

- Grade: Second Class Honours Division A (H2A)
- Scholarship: Faculty of Engineering International Undergraduate Excellence Scholarship, awarded in recognition of academic merit.
- High Distinction Coursework: Software Security (90%), Final Year Project A (85%), Software Architecture & Design (82%), Software Engineering Practice (80%).

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript, HTML, CSS, PHP

Frameworks and Libraries: React, FastAPI, Flask, Node.js, PyTorch, LangChain, FAISS, Pandas, NumPy

Data and AI: Data pipelines, ETL, data modelling, RAG pipelines, LLM integration, machine learning, computer vision, prompt engineering

Cloud and Infrastructure: Microsoft Azure, PostgreSQL, MongoDB, REST APIs, OAuth 2.0, Git, GitHub, CI/CD, Linux

Tools and Methodologies: Claude Code, Cursor, GitHub Copilot, Playwright, Tableau, Agile, code reviews, automated testing, technical documentation

WORK EXPERIENCE

AI Developer | QEERI.AI

Mar 2024 - Oct 2024

Melbourne, VIC (Industry Placement via Monash University)

- Built and deployed a production AI platform helping students navigate career pathways, using Python, LangChain and FAISS with a React frontend and FastAPI backend on Microsoft Azure, serving 100+ active users at 98% uptime.
- Diagnosed a live production failure where the platform began returning incorrect results, tracing it through the retrieval layer to a document chunking defect, rebuilding that component and improving accuracy by 40%.
- Designed and built automated data pipelines processing 500+ documents from multiple sources, reducing manual processing effort by 95% and improving data quality across the platform.
- Worked directly with stakeholders to understand the underlying problem rather than the requested feature, and translated technical findings clearly for non-technical audiences.

Computer Vision Engineer | CSIRO

Mar 2023 - Oct 2023

Melbourne, VIC (Industry Placement via Monash University)

- Trained and evaluated computer vision models in Python and PyTorch for a government research project on wearable AR hardware, preparing the training datasets and achieving 92% accuracy across 50+ categories.
- Profiled and optimised the processing pipeline running on deployed hardware, reducing system latency by 15ms and diagnosing integration failures that brought error rates below 1%.
- Produced technical documentation and presented findings to researchers and engineers from non-software backgrounds across an interdisciplinary team.

Full Stack Web Developer | Creation Export

Dec 2021 - Feb 2022 | Ongoing 2024 - 2025

- Designed, built and launched a full stack e-commerce platform using JavaScript, PHP, REST APIs and OAuth 2.0, including the relational SQL database schema handling concurrent product listings and user accounts.
- Delivered ahead of the three month deadline and have provided ongoing maintenance and feature development since.

Avionics Software Developer | Monash Uncrewed Aerial Systems

Apr 2022 - Feb 2024

Melbourne, VIC (Monash University Engineering Students Team)

- Developed autonomous flight control software within a 50-member cross-functional engineering team, contributing to a 15% increase in flight mission success rate.
- Resolved communication and telemetry protocol failures across 10+ interconnected electronic systems, reducing overall error rates by 25%.

Platform achieved Top 10% global ranking at The New Flying Competition in Hamburg, Germany, after I had moved on from the team in February 2024.

PROJECTS

Job Market Analytics Dashboard | Python, Node.js, React, Gemini API

2026

- Full stack analytics platform that aggregates job market data via the Adzuna API, tracks application pipelines and visualises hiring trends through a React dashboard with real-time metrics.
- Built end to end using Claude Code and Cursor as primary development tools, with automated data ingestion workflows in Python and Node.js.

Universal Book | React, Supabase, PostgreSQL, REST APIs

2025

Monash Honours Capstone Project

- Real-time collaborative online library built end to end, including relational PostgreSQL schema design and custom REST API endpoints.

Logistics & Resource Optimisation Engine | Python, Graph Theory, Network Flow

2025

- Formulated a constrained multi-source transit allocation system by mapping spatial locations and transportation corridors into a directed weighted graph, optimising reachability metrics under bounded distance thresholds.
- Modelled the allocation problem as a network flow graph and applied a max-flow algorithm to assign students across buses while respecting individual minimum and maximum capacity constraints within required time complexity.

CERTIFICATIONS

- HackerRank Software Engineer Certificate (May 2026)
- 100 Days of Code: Complete Python Pro Bootcamp - Udemy (Jul 2024)
- ANZ Cyber Security Management Virtual Experience (Oct 2022)
- IELTS Academic - British Council (Jan 2026)